DATASCI 406 FA 26, Quiz 2.4

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

$X \sim \text{Exp}(1)$, and given $X = x$, $Y \sim \text{Poisson}(x)$. Which code draws `n` pairs (X, Y) from this joint distribution?

1. `x <- rexp(n); y <- rpois(n, x)`
2. `x <- rexp(n); y <- rpois(n, rexp(n))`
3. `x <- rexp(n); y <- rpois(n, mean(x))`
4. `y <- rpois(n, 1); x <- rexp(n, y)`

Solution

Answer: 1. Draw X from its marginal, then each Y_i given its own X_i. Option 2 gives Y the right marginal but draws it independently of X, which a QQ plot of Y alone would not catch. Option 3 uses one rate for every Y_i. Option 4 invents a factoring the problem does not give.

Question 2 (1 point)

$\mathbf{Z} \sim N_2(\mathbf{0}, \mathbf{I}_2)$. For which matrix $\mathbf{C}$ is $\mathbf{CZ} \sim N_2\left(\mathbf{0}, \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\right)$? Hint: recall for matrix multiplication, $C = AB$ has $C_{ij} = \sum_k^n A_{ik} B_{kj}$.

1. $\begin{pmatrix} 4 & 0 \\ 2 & 2 \end{pmatrix}$
2. $\begin{pmatrix} 2 & 0 \\ 0 & \sqrt{2} \end{pmatrix}$
3. $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$
4. $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

Solution

Answer: 3. $\mathbf{CZ}$ has covariance $\mathbf{CC}^T$, and $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$. Option 1 skips the square root. Option 2 takes square roots of the variances and loses the covariance. Option 4 is `chol(Sigma)`, which is $\mathbf{C}^T$, and gives $\mathbf{CC}^T = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$.

Question 3 (1 point)

Why is the multivariate normal a useful model for several correlated quantities, such as test statistics computed on the same subjects?

1. It can represent any form of dependence between the quantities.
2. Its dependence is set entirely by Σ, and the CLT makes many statistics approximately normal.
3. Its components are independent, so each one can be drawn separately.
4. Probabilities such as $P(\max_i |Z_i| > c)$ have closed forms, so no simulation is needed.

Solution

Answer: 2. Dependence in the MVN is covariance and nothing else, so one matrix describes it, and the CLT means many estimators and test statistics are close to jointly normal. Option 1 overstates it: covariance captures only linear dependence. Option 3 holds only when Σ is diagonal. Option 4 is why we simulated: the FWER is a 10-dimensional normal integral.

Question 4 (1 point)

You want to simulate a stock's log price at the close of each trading day for the next year. Which is the best tool?

1. A Poisson process, with each arrival a trade.
2. Independent normal draws, one for each day.
3. A random walk with normal steps (Brownian motion on a daily grid).
4. A spatial Poisson process, with days as the region.

Solution

Answer: 3. Each day's log price is the previous day's plus a random change, so the process depends on the past through the most recent value. This is the model the turn-in fit to the S&P 500. Option 2 forgets yesterday's price entirely. Options 1 and 4 count events and cannot go down.

Rubric. 1 pt per question, no partial credit.